

Data Wrangling I

The questions below refer to the data frame called `day_at_cal`, with the habits of five students during a day at Cal including the day of the week (`weekday`), their study spot that day (`study_spot`), the number of total hours that they used AI in the course of their work (`ai`), and the number of steps walked, in thousands (`steps_k`). To check your answers in R, you can copy and paste the code that creates this data frame from the last slide from class.

```
day_at_cal
```

```
# A tibble: 6 × 4
  weekday study_spot ai_hours steps_k
  <ord>    <fct>      <dbl>   <dbl>
1 Mon     Moffitt      1.2     8.5
2 Mon     Doe          0.4     6.2
3 Tue     Cory         2.1     9.1
4 Tue     MLK           1       7
5 Wed     CITRIS        0.8     5.8
6 Wed     Soda         1.6    10.3
```

Write a single `dplyr` command that will produce each of the following.

1. A data frame containing the 2nd and 5th students.
2. A data frame without the 3rd student.
3. A data frame with only the AI hours column.
4. A data frame with only the factor columns¹.
5. A data frame containing only the students with at least one hour of AI use.
6. A data frame with students who study in either Moffitt or Doe.
7. A data frame with a fifth column called `ai_min` with the AI use in minutes.
8. A data frame with a fifth (logical) column called `active_day` where the number of steps exceeds 8,000.

¹There is a clever way to do this that avoids having to type the names of the columns that are factors. See the help file for `select()`, particularly about predicate functions.

9. The same data frame sorted in descending order of number of hours of AI.
10. The same data frame sorted according to the day of the week. Within rows with the same day of the week, sort in descending order of the number of steps.
11. A data frame with the overall mean hours of AI use. Name that statistic `mean_ai`.
12. A data frame with the minimum steps walked (`min_steps`), the maximum steps walked (`max_steps`), and the standard deviation of steps walked (`sd_steps`).
13. Skim the documentation for the `tibble` package (<https://tibble.tidyverse.org/>) then write down three ways in which tibbles behave differently than data frames.
14. If you're interested in reading more about the design philosophy behind the Tidyverse, read *The Tidy Tools Manifesto* by Hadley Wickham, <https://cran.r-project.org/web/packages/tidyverse/vignettes/manifesto.html>, (it's just two pages).

Data Wrangling II
Coming soon!